

The Problem

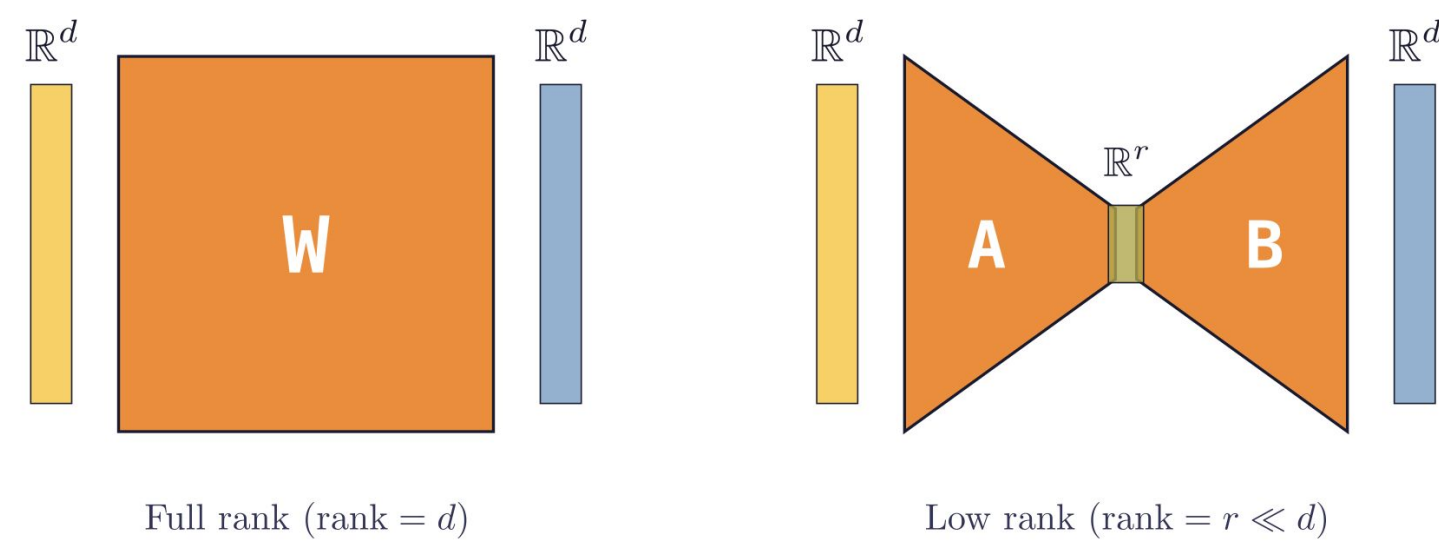
GPT-3 has
175B parameters

How do we fine-tune this?

Full fine-tuning would take $175B \times 2 \text{ bytes} = \sim 350 \text{ GB}$ per task...

The Idea

Do we need to fine-tune all parameters?

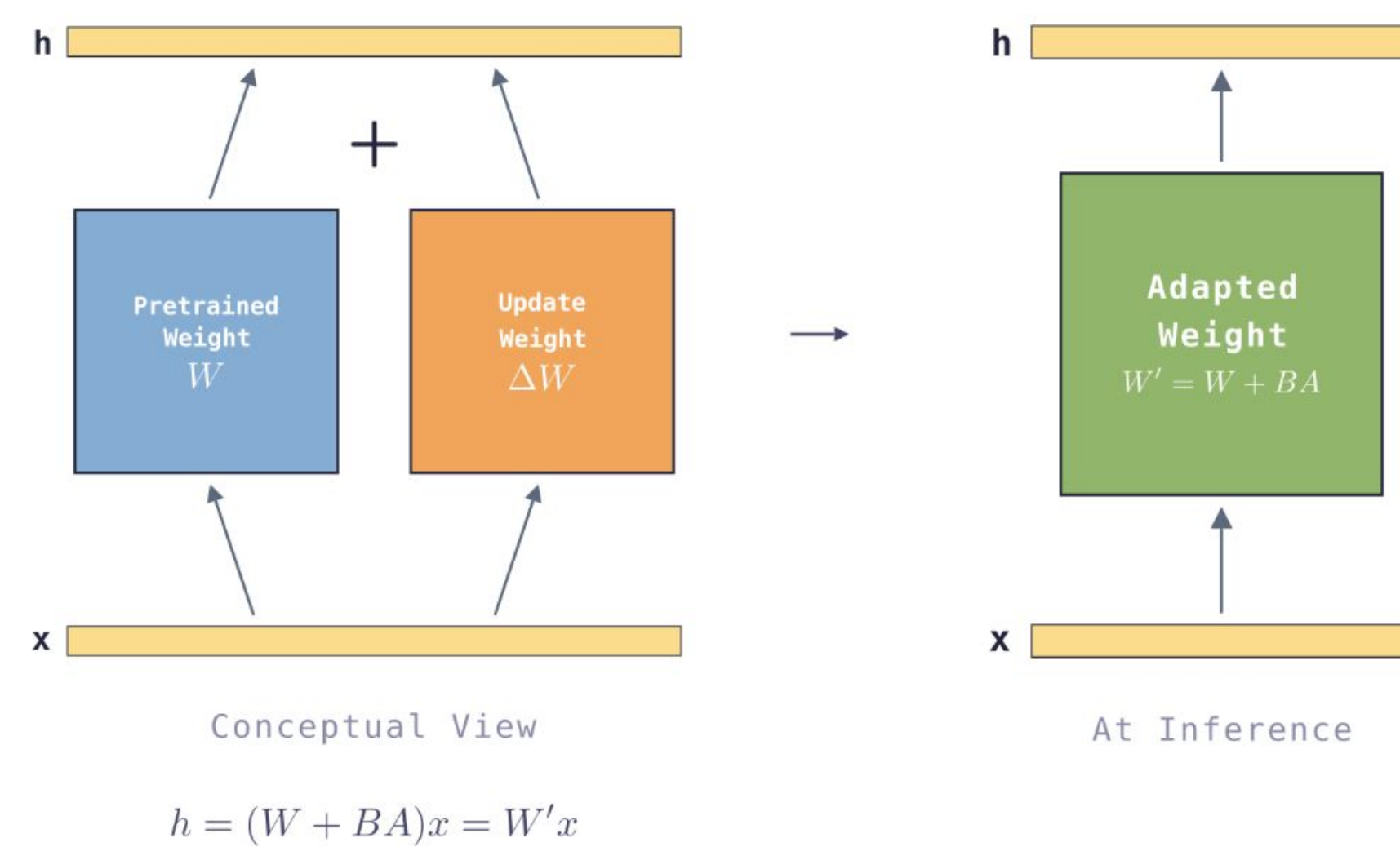


We **bottleneck** our parameter updates.

Instead of update matrix ΔW , we learn smaller matrices A and B.

These combine to form $\Delta W = B \times A$

Rank (r) defines the width of the information bottleneck.



At inference time, we expand $B \times A$ and add the update weights.
LoRA adds no inference latency.

By decomposing the update into A ($r \times d$) and B ($d \times r$),
We decrease parameters from d^2 to $2rd$

Reproducing LoRA

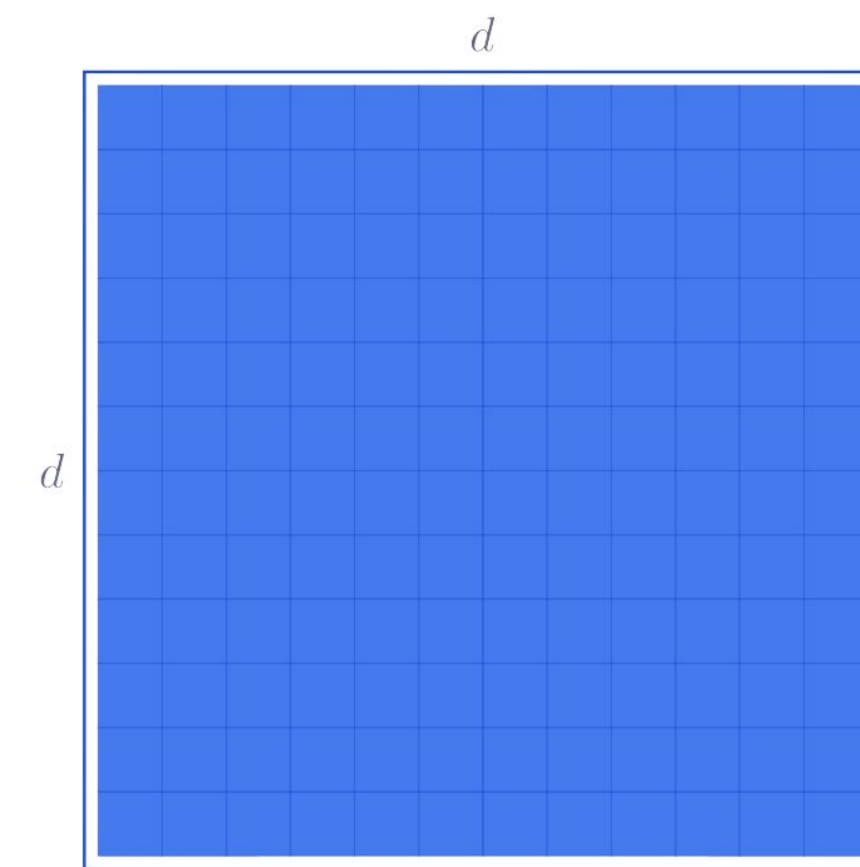
Low-Rank Adaptation of RoBERTa-base on GLUE Benchmarks

Lucas He

Aden Zhao

Full Fine-Tuning

100% of parameters trainable

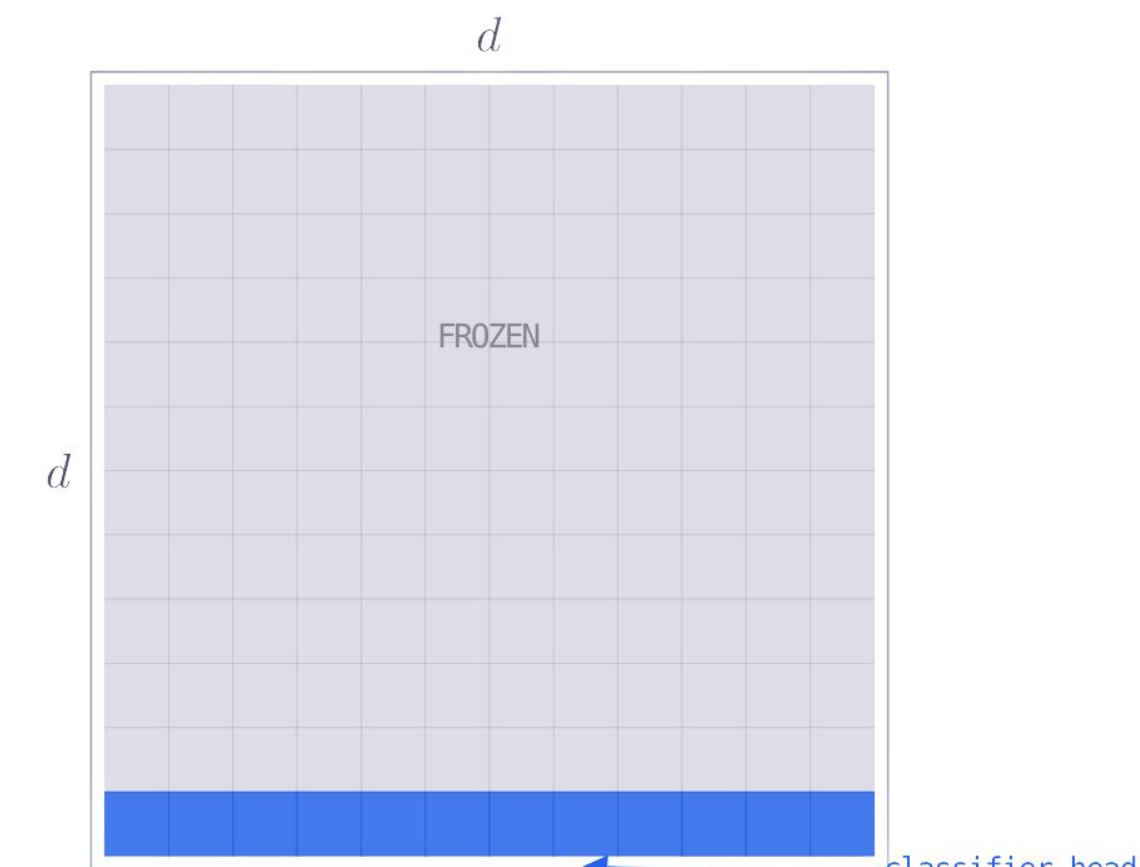


88.2% Accuracy

350 GB per task • $d \times d$ parameters

Head-Only Fine-Tuning

Only the classifier head is trainable



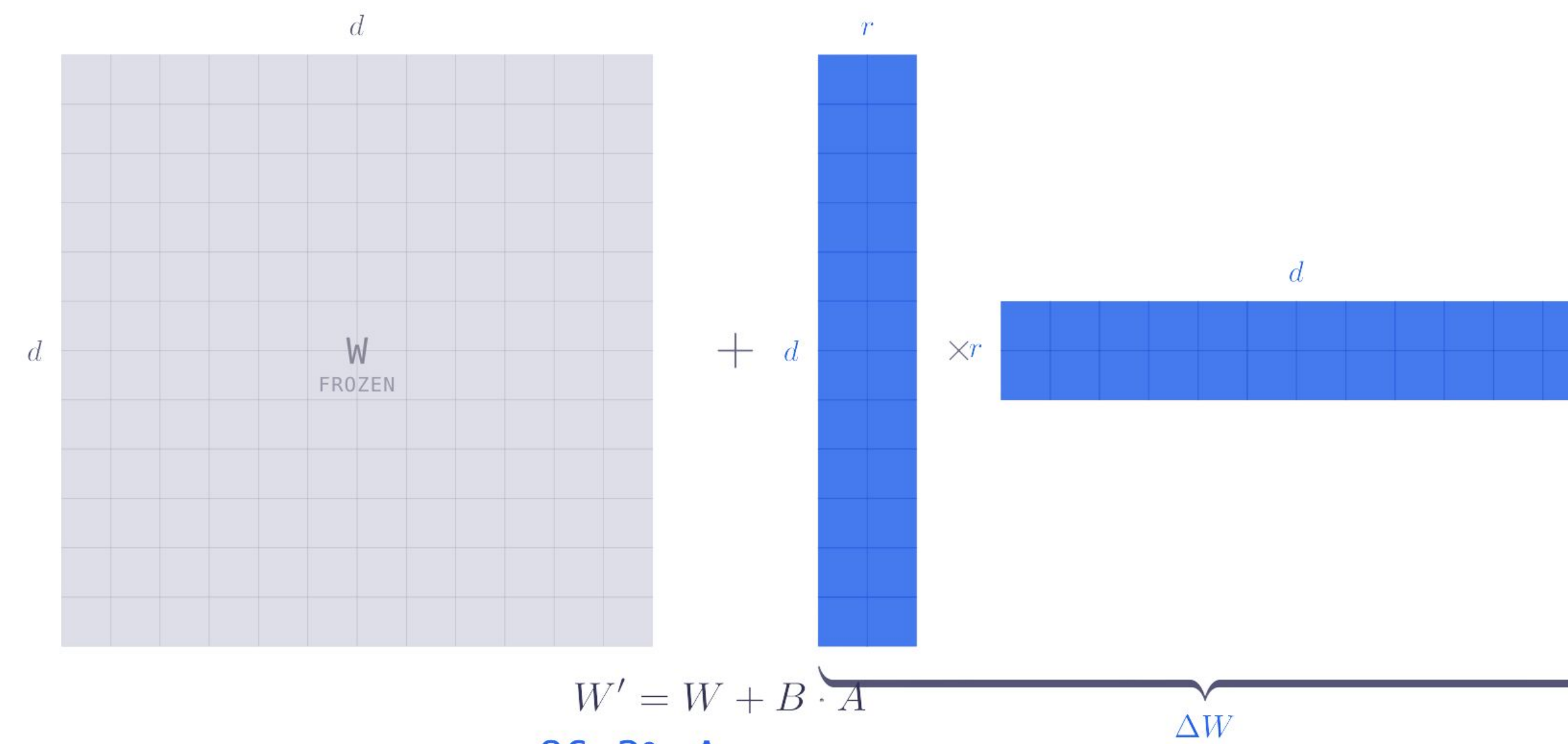
68.4% Accuracy

Minimal storage • Performance collapse

VS

LoRA

Low-Rank Adaptation • 0.7% parameters trainable



86.3% Accuracy

140x fewer params • 35 MB checkpoint

Which strategy would you use?



Our Results



LoRA Paper

Benchmark

Paraphrase Classification

- S1 Wei-Chiu said the experiment was a success and that the results exceeded expectations.
- S2 According to Wei-Chiu, the experiment succeeded beyond what they had originally hoped for.

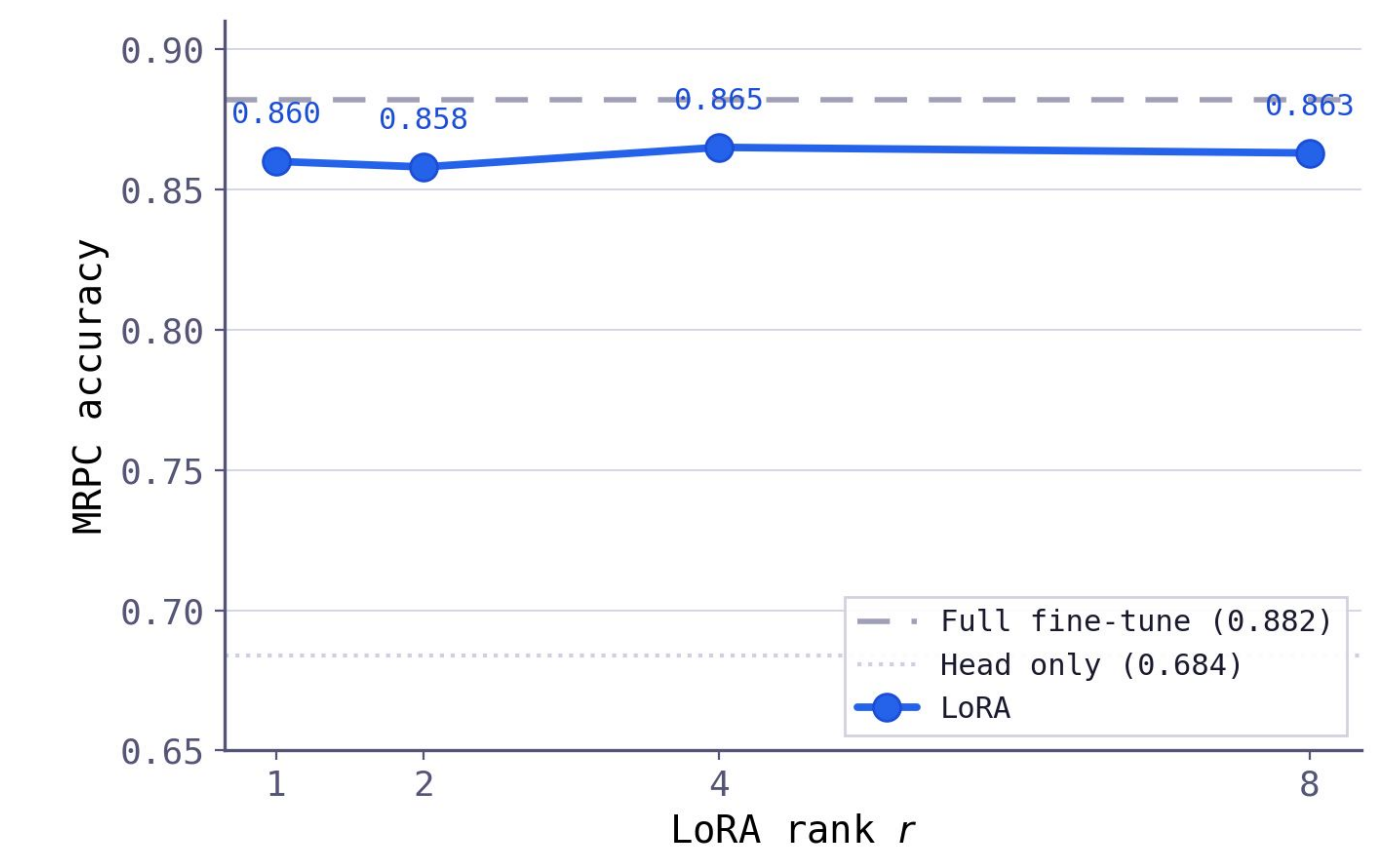
✓ PARAPHRASE

- S1 Kilian presented the findings at the conference in Berlin last Tuesday.
- S2 Lucas reviewed the findings before they were submitted to the journal on Friday.

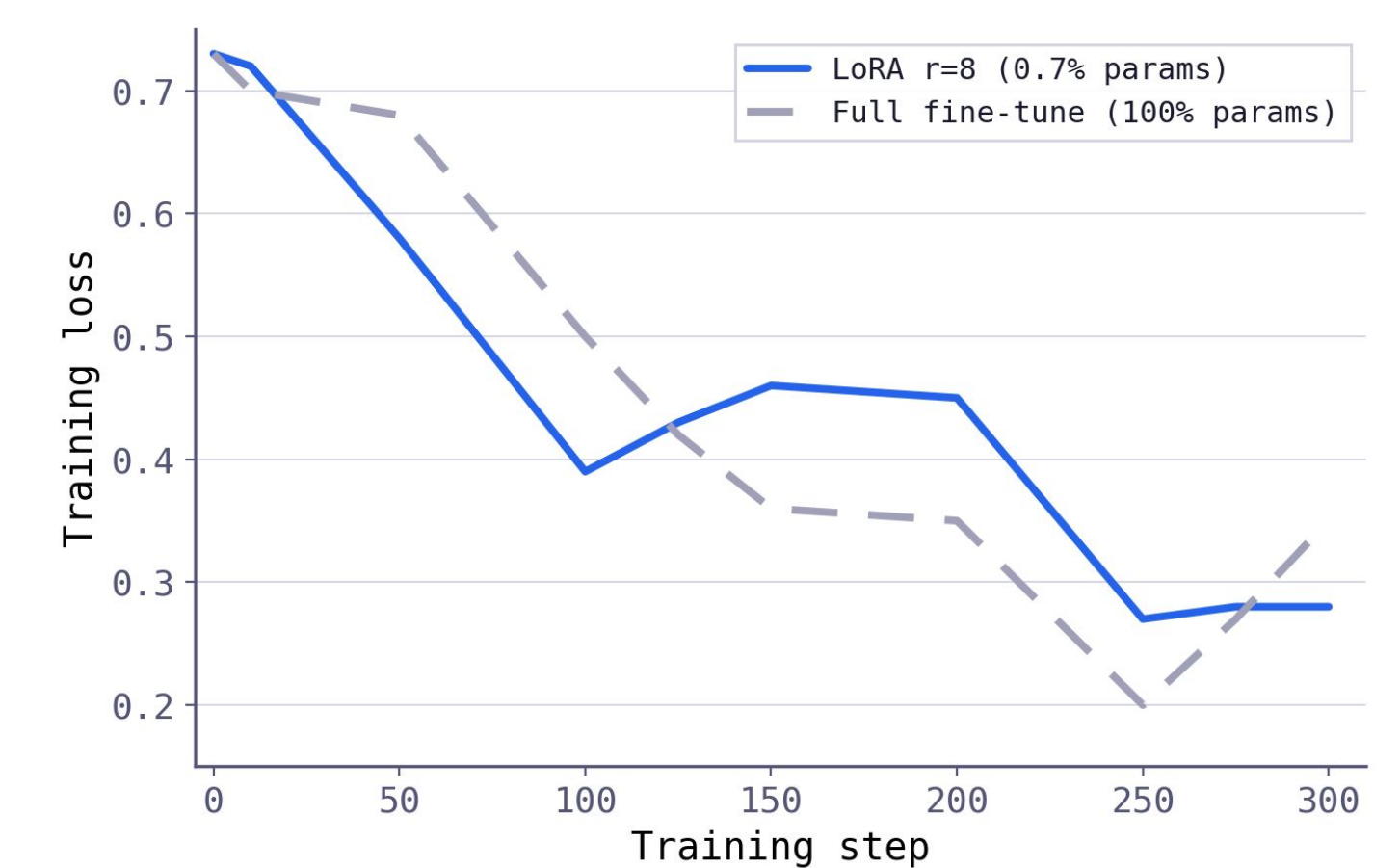
✗ NOT PARAPHRASE

Results

LoRA is almost as accurate at a fraction of the cost



Same trajectory at a fraction of the params.



Models close to the origin work just as well.

LoRA as a Generalization of Full Fine-Tuning

